AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 22 · STUDENT LESSON

Queues and asynchronous work

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Caching and stale information. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of queues and asynchronous work. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **queue**s and asynchronous work.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of **queue**s and asynchronous work

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Caching and stale information. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain **queue**s and asynchronous work using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Queues and asynchronous work case. Separate observations, given facts and assumptions before choosing an explanation.

Queues separate acceptance from completion. Work may be redelivered, so consumers must preserve operation IDs and deduplicate committed effects.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Queues and asynchronous work. Connect each labelled input, deciding step and result.

A **queue**d job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.

Queues and asynchronous work — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Worker commits job J then crashes before queue acknowledgement. J is delivered again.	A queued job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.	Consult durable job status; return committed result and acknowledge without repeating the effect. Delivery count is not effect count.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Queues and asynchronous work. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: The worker crashes before producing checked **output**.

Predicted result

Prediction: The job is failed or awaiting a controlled retry, not succeeded.

Why the result follows

Reason: Receipt is not **evidence** that conversion finished.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Queues and asynchronous work to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Queue	A store of work awaiting processing
Acknowledgement	A signal allowing delivery completion
Redelivery	Receiving a previously delivered job again

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Queues and asynchronous work, predict a step, then compare with its explanation.

Worker commits job J then crashes before **queue acknowledgement**. J is delivered again.

Consult durable job status; return committed result and acknowledge without repeating the effect. Delivery count is not effect count.

Worked explanation · Queues and asynchronous work

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Queues and asynchronous work

Queued is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.

The facts we will use

Supplied facts

States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.

AI ACADEMY / AI-6/22

Queues and asynchronous work

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Accept request

J12 acknowledges receipt, not completed conversion.



2. Start worker

The worker marks J12 running before processing.



3. Publish checked result

Only a checked output permits succeeded status.

CHANGE ONE THING

The worker crashes before producing checked output.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Accept request	J12 acknowledges receipt, not completed conversion.
2. Start worker	The worker marks J12 running before processing.
3. Publish checked result	Only a checked output permits succeeded status.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Asynchronous work separates submission from completion. Honest status and a job identifier let users track that difference.

What this example does not establish

Keep the stated limit

Queue delivery and duplicate handling need their own policies; this sequence does not guarantee completion.

Keep the stated limit

Read the labelled picture: Explain what the **queue**d J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?

Keep the stated limit

Change one thing: The worker crashes before producing checked **output**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Queues and asynchronous work claim and explain a repair using the supplied evidence.

Queue ordering and exactly-once properties must be declared, not assumed from a simple mock.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: **Queue** ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Queues and asynchronous work with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**Queue**s and asynchronous work". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Queue**s and asynchronous work. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Explain what the **queue**d J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Queues and asynchronous work. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Queues and asynchronous work. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: **Queue** ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled **queue**s and asynchronous work **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **queue**s and asynchronous work.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

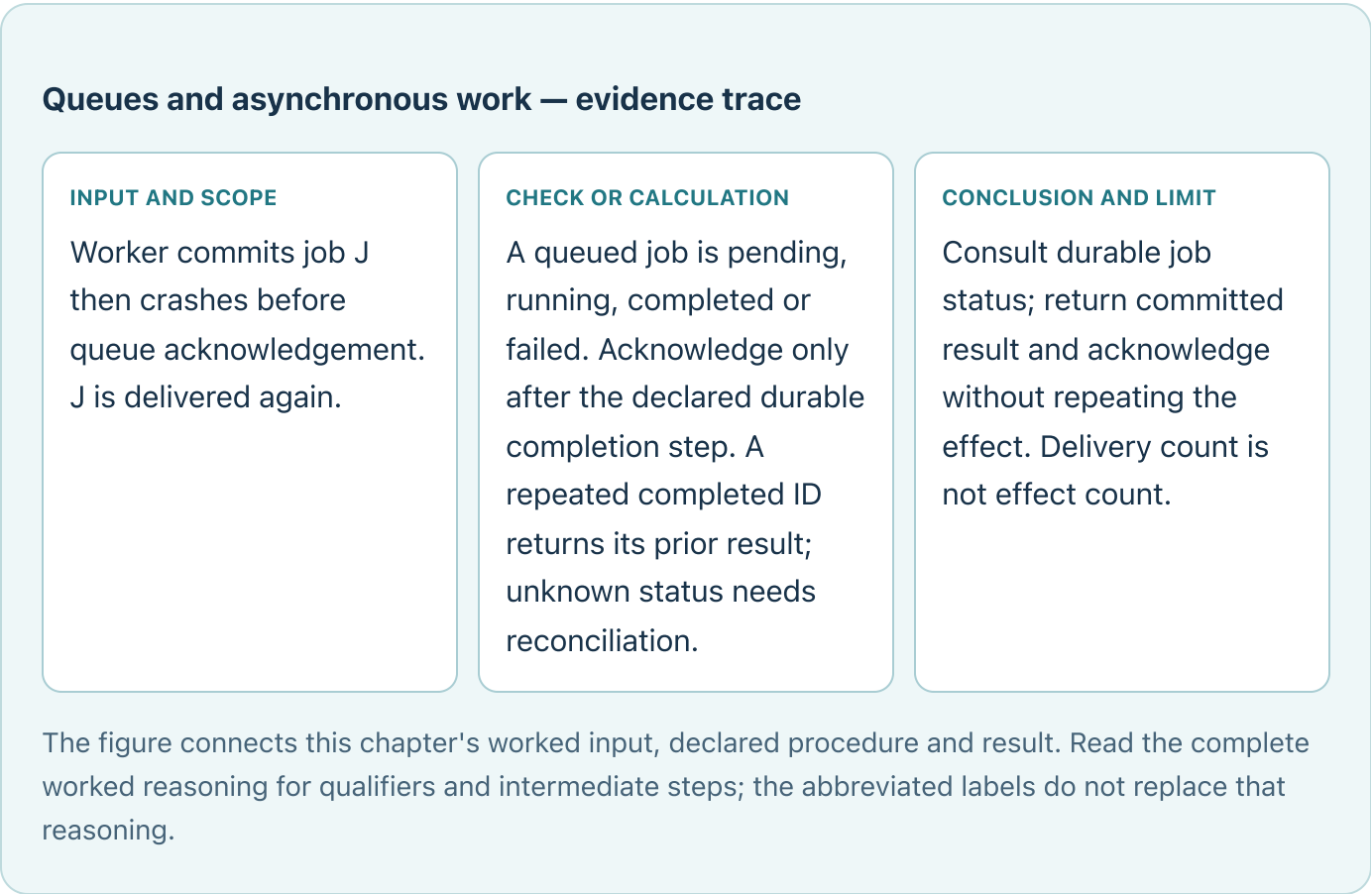
Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Queues and asynchronous work case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

> Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may

success be claimed for all accepted?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Queues and asynchronous work, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: **Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.

By the end, I can	How I will show it
Explain the mechanism and terminology of queues and asynchronous work	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Queues and asynchronous work without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **queue**s and asynchronous work

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Queued is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only a checked **output** permits succeeded status.

1. Start worker

2. Publish checked result

3. Accept request

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Publish checked result Only a checked **output** permits succeeded status. Asynchronous work separates submission from completion.

Honest status and a job identifier let users track that difference.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The worker crashes before producing checked **output**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The job is failed or awaiting a controlled retry, not succeeded.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The job is failed or awaiting a controlled retry, not succeeded. Receipt is not **evidence** that conversion finished.

Repair a missing building block

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version-5 request return, and what permits a later version-5 cache hit?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

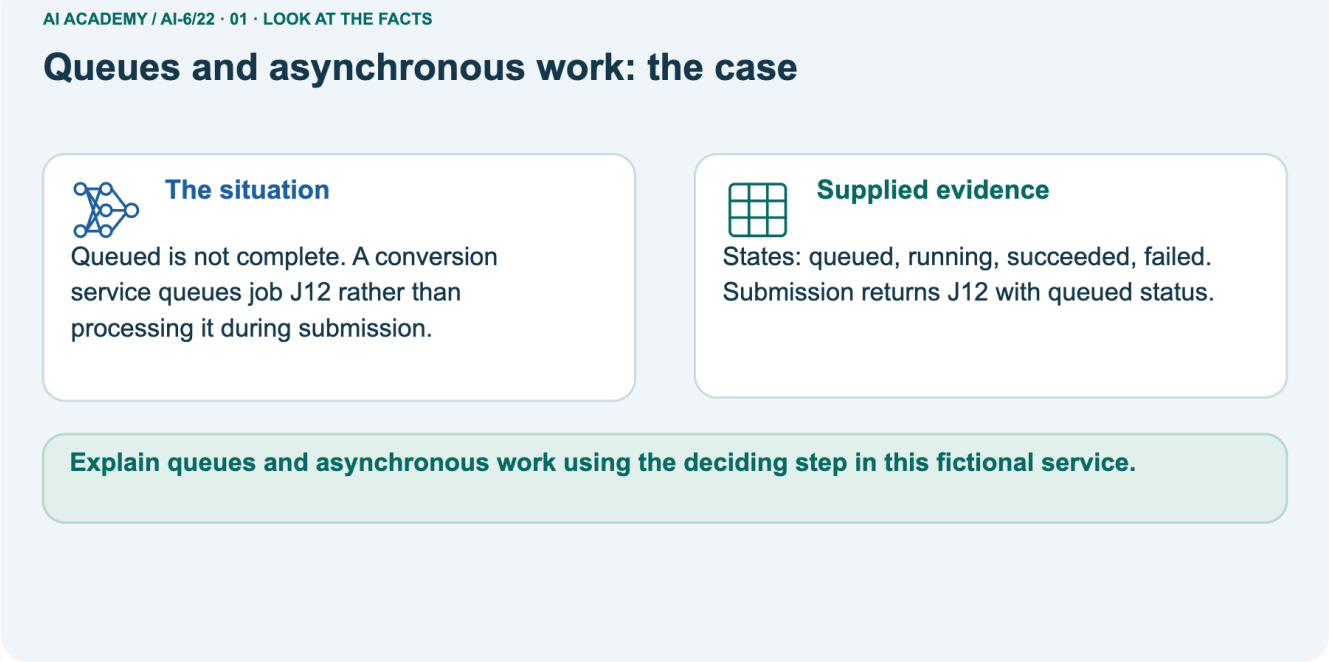
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Queues and asynchronous work graphic using its labels, deciding evidence and relevant limitation.

01. Queues and asynchronous work: the case

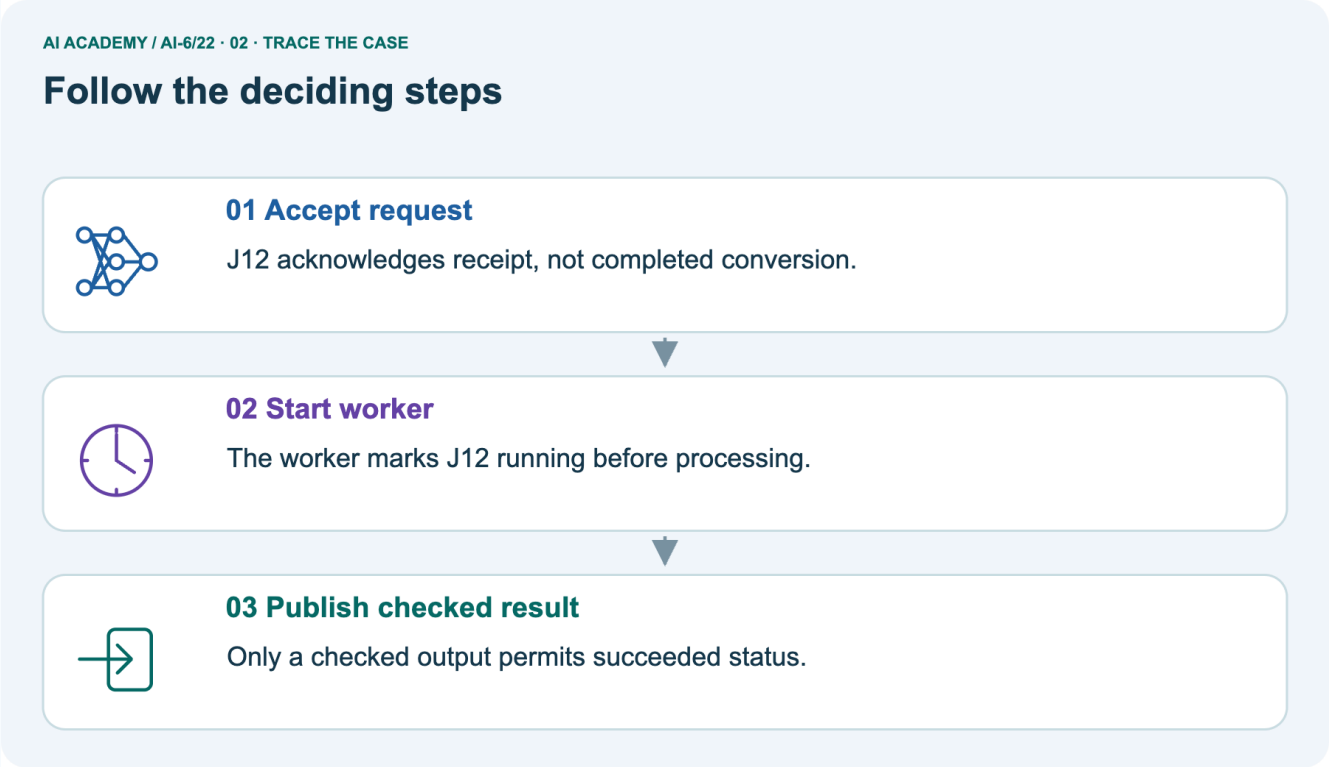


Follow the labelled picture with the matching explanation. Open the vector diagram

Queued is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission. Supplied facts: States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

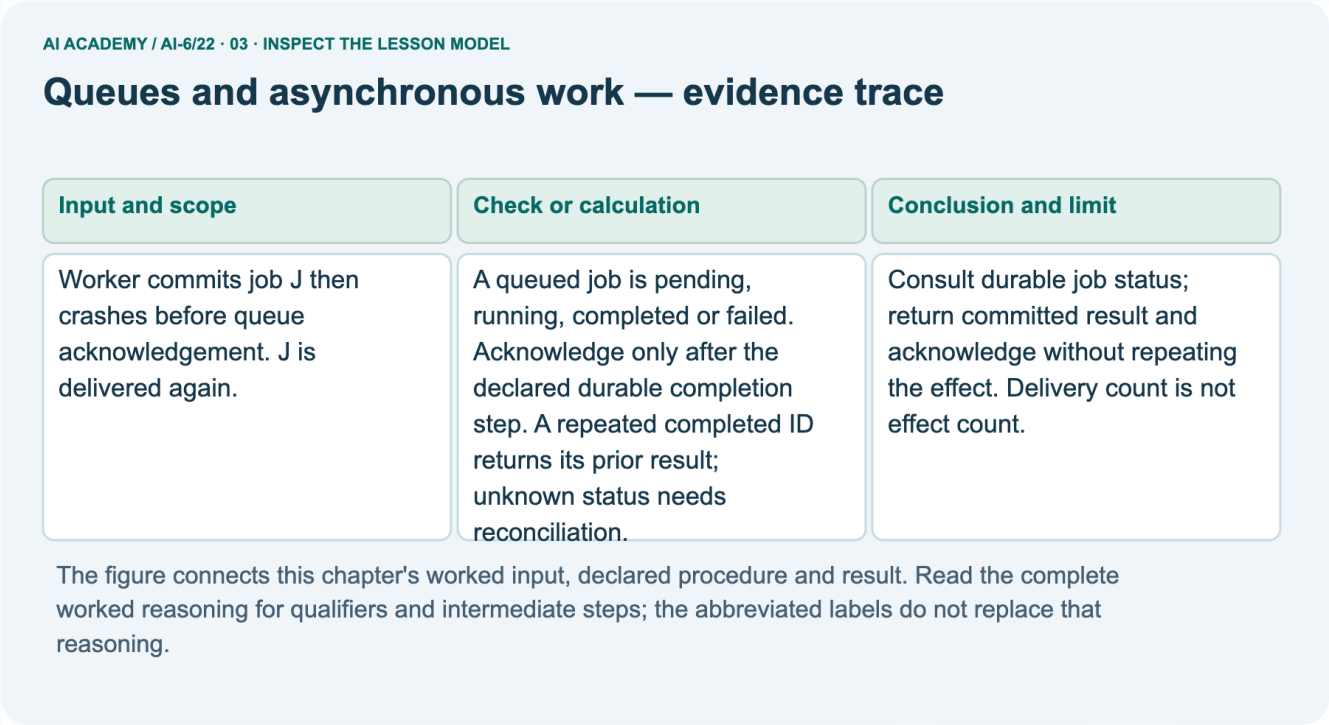


Follow the labelled picture with the matching explanation. Open the vector diagram

Accept request: J12 acknowledges receipt, not completed conversion. → Start worker: The worker marks J12 running before **process**ing. → Publish checked result: Only a checked **output** permits succeeded status.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Queues and asynchronous work — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

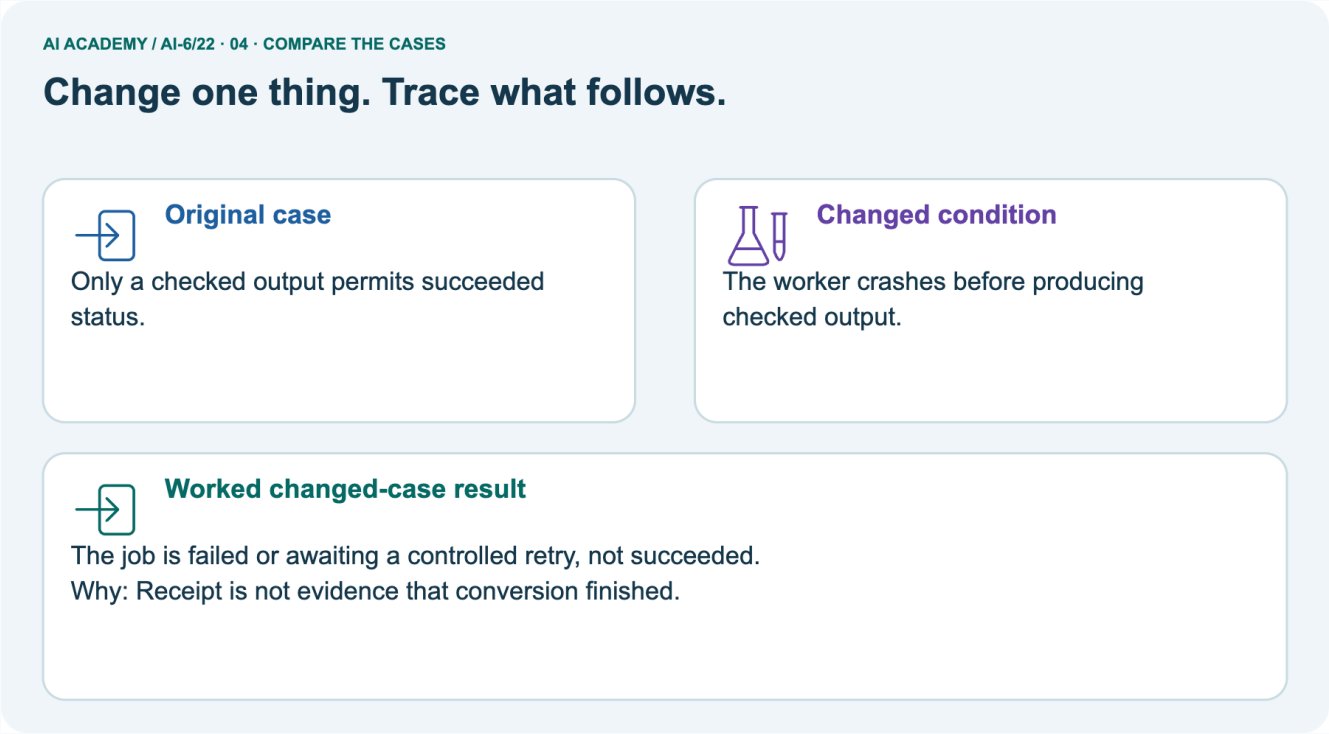
Queues and asynchronous work — **evidence** trace. Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. | A **queue**d job is pending, running, completed or failed.

Acknowledge only after the declared durable completion step.

A repeated completed ID returns its prior result; unknown status needs reconciliation. | Consult durable job status; return committed result and acknowledge without repeating the effect. Delivery count is not effect count.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



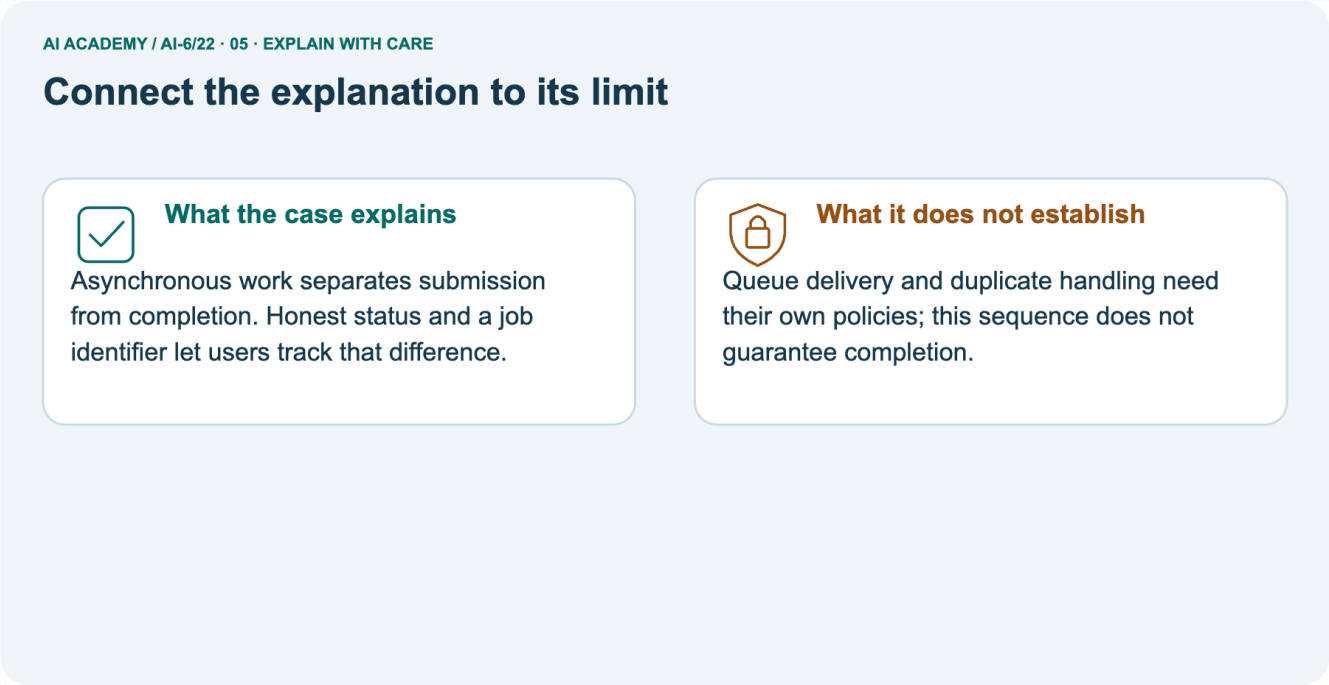
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only a checked **output** permits succeeded status. Changed condition: The worker crashes before producing checked **output**. Expected result: The job is failed or awaiting a controlled retry, not succeeded.

Reason: Receipt is not **evidence** that conversion finished.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



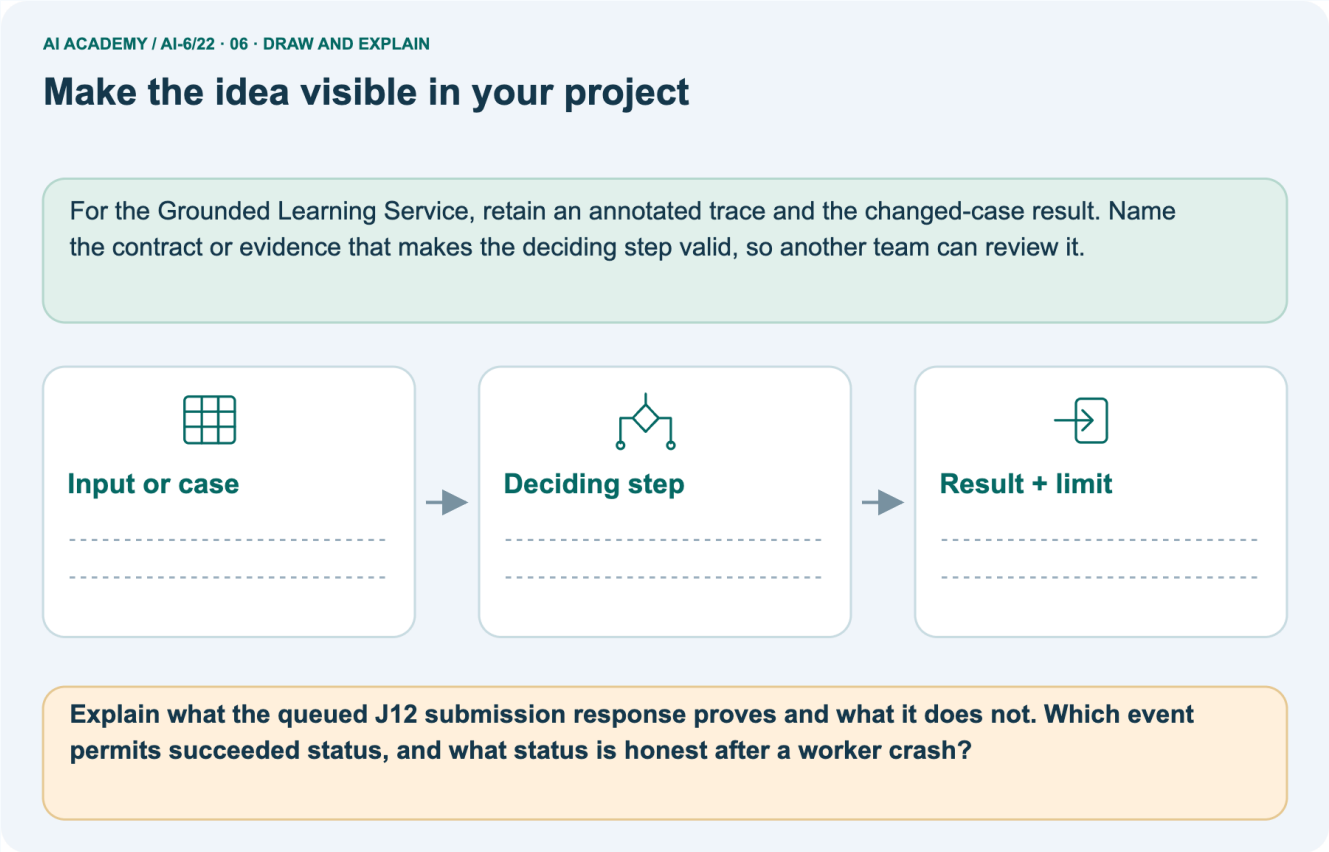
Follow the labelled picture with the matching explanation. Open the vector diagram

Asynchronous work separates submission from completion. Honest status and a job identifier let users track that difference.

Limit: **Queue** delivery and duplicate handling need their own policies; this sequence does not guarantee completion.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain what the **queue**d J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.